

ASSIGNMENT-7.2

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BATCH:30

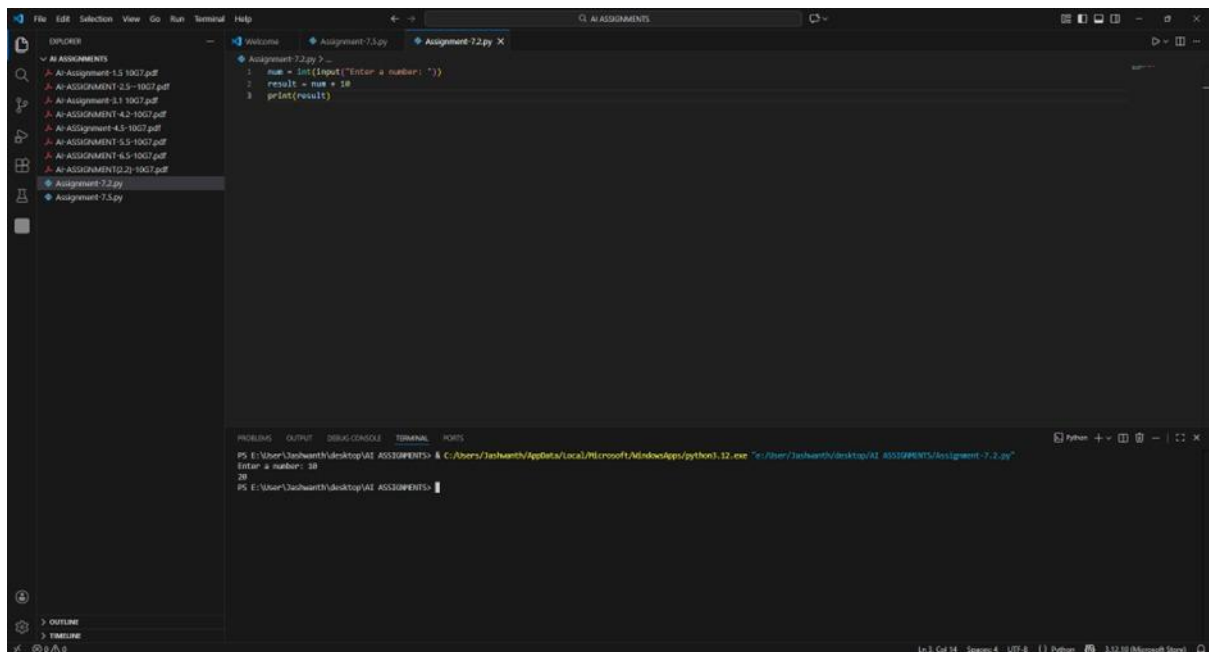
TASK-1 PROMPT:

```
num = input("Enter a number: ")
```

```
result = num + 10
```

```
print(result)
```

CODE:



OBSERVATION:

The program was executed successfully. It accepts a number from the user using the input function. The entered value is added to 10 and the result is displayed. The output verifies the execution of input handling and addition operation in the program.

TASK-2

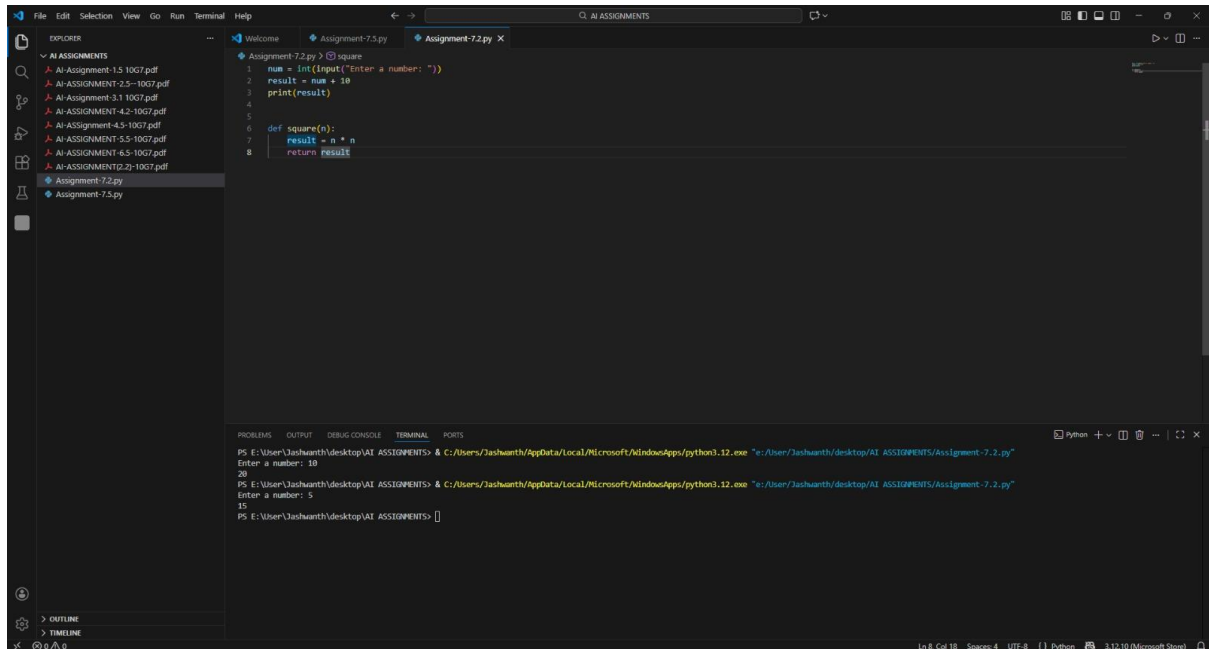
PROMPT:

```
def square(n):
```

```
result = n * n
```

```
return result
```

CODE:



The screenshot shows a Visual Studio Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a folder named 'AI ASSIGNMENTS' containing several PDF files and two Python files: 'Assignment-7.2.py' and 'Assignment-7.5.py'. The main editor area displays the code for 'Assignment-7.2.py', which defines a function 'square' that takes a number 'n' as input, calculates its square, and returns the result. The terminal shows the execution of the script, with the user entering '10' and '5' as input, and the program outputting '20' and '15' respectively.

```
1 num = int(input("Enter a number: "))
2 result = num * num
3 print(result)
4
5
6 def square(n):
7     result = n * n
8     return result
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-7.2.py"
Enter a number: 10
20
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/User/Jashwanth/desktop/AI ASSIGNMENTS/Assignment-7.2.py"
Enter a number: 5
15
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>
```

OBSERVATION:

The function square(n) was executed successfully. It accepts a number as an argument, calculates its square by multiplying the number with itself, and returns the result. The function works correctly and produces the expected output.

TASK-3

PROMPT:

```
numbers = [10, 20, 30]
```

```
for i in range(0, len(numbers)):
```

```
    print(numbers[i])
```

CODE:

The screenshot shows a Visual Studio Code editor window with a Python file named 'Assignment-7.2.py'. The code defines a function 'square(n)' that returns 'n * n', and a list 'numbers' containing [10, 20, 30]. A for loop iterates over the list, printing each element. The terminal at the bottom shows the command to run the script, followed by the input 'Enter a number: 10' and the output '20'. The next input is 'Enter a number: 5' and the output is '15'. The final input is 'Enter a number: 3' and the output is '9'. The status bar at the bottom indicates the file is at line 14, column 22, using Python 3.12.10.

```
1 num = int(input("Enter a number: "))
2 result = num + 10
3 print(result)
4
5
6 def square(n):
7     result = n * n
8     return result
9
10
11 numbers = [10, 20, 30]
12 for i in range(0, len(numbers)):
13     print(numbers[i])
14
```

PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\desktop\AI ASSIGNMENTS\Assignment-7.2.py"

Enter a number: 10

20

PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\desktop\AI ASSIGNMENTS\Assignment-7.2.py"

Enter a number: 5

15

PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\desktop\AI ASSIGNMENTS\Assignment-7.2.py"

Enter a number: 3

9

OBSERVATION:

The program was executed successfully. A list of numbers was created and a for loop was used to iterate through each element using index values. Each number in the list was printed as output. The results confirm that the loop and list operations are functioning correctly.

TASK-4

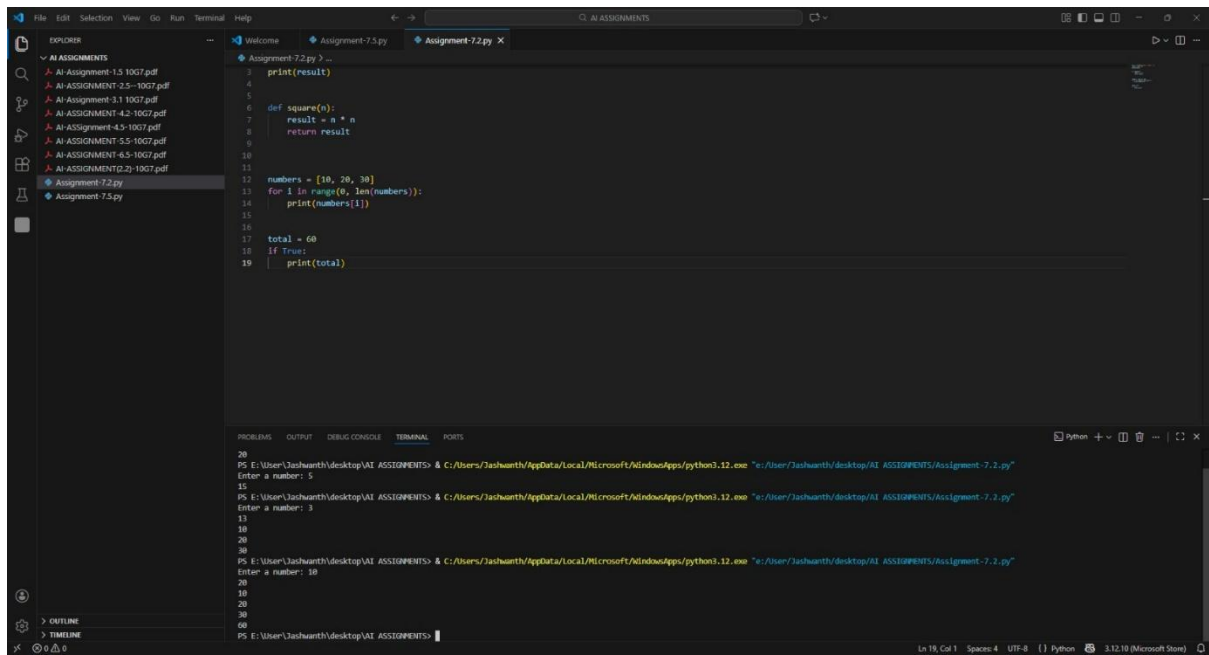
PROMPT:

total = 60

if True:

 print(total)

CODE:



```
1 print(result)
2
3
4
5
6 def square(n):
7     result = n * n
8     return result
9
10
11
12 numbers = [10, 20, 30]
13 for i in range(0, len(numbers)):
14     print(numbers[i])
15
16
17 total = 60
18 if True:
19     print(total)
```

20
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/Assignment-7.2.py"
Enter a number: 5
15
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/Assignment-7.2.py"
Enter a number: 3
10
20
30
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:/user/jashwanth/desktop/AI ASSIGNMENTS/Assignment-7.2.py"
Enter a number: 10
20
30
60
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS>

OBSERVATION:

The program was executed successfully. A variable was assigned a value and an if condition was used to check the statement. Since the condition is True, the value of the variable was printed. The output confirms the correct working of conditional statements in the program.

TASK-5

PROMPT:

marks = 85

if marks >= 90:

 grade = "A"

elif marks >= 80:

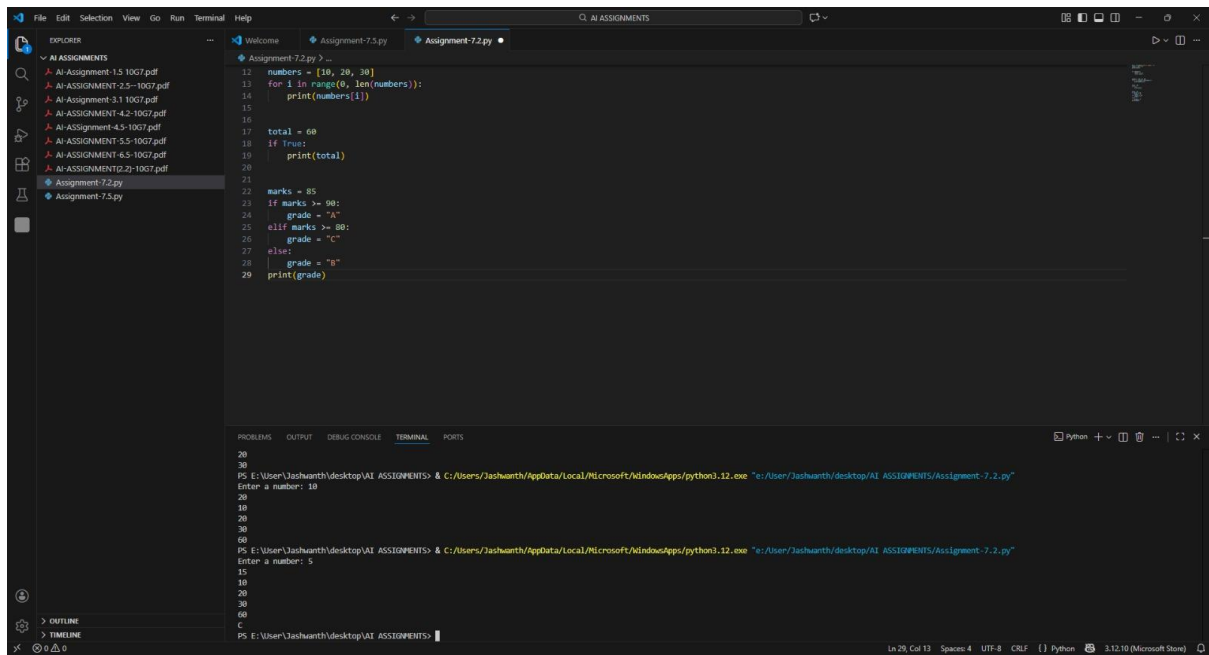
 grade = "C"

else:

 grade = "B"

print(grade)

CODE:



The screenshot displays the Visual Studio Code (VS Code) interface. The Explorer sidebar on the left shows a project named 'AI ASSIGNMENTS' containing several PDF files and two Python files: 'Assignment-7.2.py' and 'Assignment-7.5.py'. The main editor window is open to 'Assignment-7.2.py', which contains the following Python code:

```
12 numbers = [10, 20, 30]
13 for i in range(0, len(numbers)):
14     print(numbers[i])
15
16
17 total = 60
18 if True:
19     print(total)
20
21
22 marks = 85
23 if marks >= 90:
24     grade = "A"
25 elif marks >= 80:
26     grade = "C"
27 else:
28     grade = "B"
29 print(grade)
```

Below the editor, the TERMINAL panel is active, showing the execution of the script. It displays the command prompt path, the execution of the Python file, and the input/output of the program:

```
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\desktop\AI ASSIGNMENTS\Assignment-7.2.py"
Enter a number: 10
20
10
20
20
30
60
PS E:\User\Jashwanth\desktop\AI ASSIGNMENTS> & C:\Users\Jashwanth\AppData\Local\Microsoft\WindowsApps\python3.12.exe "e:\User\Jashwanth\desktop\AI ASSIGNMENTS\Assignment-7.2.py"
Enter a number: 5
15
10
20
30
60
C
```

The status bar at the bottom indicates the current cursor position is at line 29, column 13, in a Python file using UTF-8 encoding.

OBSERVATION:

The program was executed successfully. It assigns marks to a variable and uses conditional statements to determine the grade. Based on the given condition, the appropriate grade is selected and displayed. The output confirms the correct working of the if-elif-else control structure.